
CHAPTER 4

Probability

Often in life we are confronted by our own ignorance. Whether we are pondering tonight's traffic jam, tomorrow's weather, next week's stock prices, an upcoming election, or where we left our hat, often we do not know an outcome with certainty. Instead, we are forced to guess, to estimate, to hedge our bets.

Probability is the science of uncertainty. It provides precise mathematical rules for understanding and analyzing our own ignorance. It does not tell us tomorrow's weather or next week's stock prices; rather, it gives us a framework for working with our limited knowledge and for making sensible decisions based on what we do and do not know.

To say there is a 40% chance of rain tomorrow is not to know tomorrow's weather. Rather, it is to know what we do not know about tomorrow's weather.

In this chapter, we will develop a more precise understanding of what it means to say there is a 40% chance of rain tomorrow. We will learn how to work with ideas of randomness, probability, expected value, prediction, estimation, etc., in ways that are

sensible and mathematically clear.

At the beginning of the twentieth century, Russians such as Andrei Andreyevich Markov, Andrey Nikolayevich Kolmogorov, and Pafnuty L. Chebyshev (and American Norbert Wiener) developed a more formal mathematical theory of probability. In the 1950s, Americans William Feller and Joe Doob wrote important books about the mathematics of probability theory. They popularized the subject in the western world, both as an important area of pure mathematics and as having important applications in physics, chemistry, and later in computer science, economics, and finance. Probability theory also plays a key role in many important applications of science and technology. For example, the design of a nuclear reactor must be such that the escape of radioactivity into the environment is an extremely rare event. Of course, we would like to say that it is categorically impossible for this to ever happen, but reactors are complicated systems, built up from many interconnected subsystems, each of which we know will fail to function properly at some time. Furthermore, we can never definitely say that a natural event like an earthquake cannot occur that would damage the reactor sufficiently to allow an emission. The best we can do is try to quantify our uncertainty concerning the failures of reactor components or the occurrence of natural events that would lead to such an event. This is where probability enters the picture. Using probability as a tool to deal with the uncertainties, the reactor can be designed to ensure that an unacceptable emission has an extremely small probability — say, once in a billion years—of occurring.

In this Chapter we introduce the basic concepts underlying probability theory. We begin with the sample space, which is the set of possible outcomes.

4.1 Sample Spaces and Events

The **sample space** Ω is the set of possible outcomes of an experiment. Points ω in Ω are called **sample outcomes**, **realizations**, or **elements**. Subsets of Ω are called **Events**.

Example. If we toss a coin twice then $\Omega = \{HH, HT, TH, TT\}$. The event that the first toss is heads is $A = \{HH, HT\}$.

Example. Let ω be the outcome of a measurement of some physical quantity, for example, temperature. Then $\Omega = \mathbb{R}$. One could argue that taking $\Omega = \mathbb{R}$ is not accurate since temperature has a lower bound. But there is usually no harm in taking the sample space to be larger than needed. The event that the measurement is larger than 10 but less than or equal to 23 is $A =]10, 23]$.

Example. If we toss a coin forever, then the sample space is the infinite set

$$\Omega = \{\omega = (\omega_1, \omega_2, \omega_3, \dots) : \omega_i \in \{H, T\}\}.$$

Let E be the event that the first head appears on the third toss. Then

$$E = \{(\omega_1, \omega_2, \omega_3, \omega_4, \dots, \omega_i, \dots) : \omega_1 = \omega_2 = T, \omega_3 = H, \forall i \geq 4, \omega_i \in \{H, T\}\}$$

Given an event A , let $A^c = \{\omega \in \Omega : \omega \notin A\}$ denote the **complement of A**. Informally, A^c can be read as “**not A**.” The complement of Ω is the empty set $\emptyset$. The union of events A and B is defined

$$A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B \text{ or } \omega \in \text{ both}\}$$

which can be thought of as “ A or B .” If $A_1, A_2, \dots$ is a sequence of sets then

$$\cup_{i=1}^{\infty} A_i = \{\omega \in \Omega : \omega \in A_i \text{ for at least one } i\}.$$

The intersection of A and B is

$$A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}$$

read “ A and B .” Sometimes we write $A \cap B$ as AB or (A, B) . If $A_1, A_2, \dots$ is a sequence of sets then

$$\cap_{i=1}^{\infty} A_i = \{\omega \in \Omega : \omega \in A_i \text{ for all } i\}.$$

The set difference is defined by

$$A - B = \{\omega : \omega \in A, \omega \notin B\}.$$

. If every element of A is also contained in B we write $A \subset B$ or, equivalently, $B \supset A$. If A is a finite set, let $|A|$ denote the number of elements in A . See the following table for a summary.

Summary of Terminology	
Ω	sample space
ω	outcome (point or element)
A	event (subset of Ω)
A^c	complement of A (not A)
$A \cup B$	union (A or B)
$A \cap B$ or AB	intersection (A and B)
$A - B$	set difference (ω in A but not in B)
$A \subset B$	set inclusion
$\emptyset$	null event (always false)
Ω	true event (always true)

The complement of $A \cup B$, namely, the set of elements that are in neither A nor B . So

we immediately have

$$(A \cup B)^c = A^c \cap B^c.$$

Similarly, we can show that

$$(A \cap B)^c = A^c \cup B^c,$$

namely, the subset of elements that are not in both A and B is given by the set of elements not in A or not in B .

We say that $A_1, A_2, \dots$ are **disjoint** or **are mutually exclusive** if $A_i \cap A_j = \emptyset$ whenever $i \neq j$. For example, $A_1 = [0, 1[, A_2 = [1, 2[, A_3 = [2, 3[, \dots$ are disjoint. A **partition** of Ω is a sequence of disjoint sets $A_1, A_2, \dots$ such that $\cup_{i=1}^{\infty} A_i = \Omega$. Given an event A , define the indicator function of A by

$$I_A(\omega) = I(\omega \in A) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \notin A \end{cases}$$

A sequence of sets $A_1, A_2, \dots$ is **monotone increasing** if $A_1 \subset A_2 \subset \dots$ and we define $\lim_{n \rightarrow \infty} A_n = \cup_{i=1}^{\infty} A_i$. A sequence of sets $A_1, A_2, \dots$ is **monotone decreasing** if $A_1 \supset A_2 \supset \dots$ and then we define $\lim_{n \rightarrow \infty} A_n = \cap_{i=1}^{\infty} A_i$. In either case, we will write $A_n \rightarrow A$.

Example. Let $\Omega = \mathbb{R}$ and let $A_i = [0, \frac{1}{i}[$ for $i = 1, 2, \dots$. Then $\cup_{i=1}^{\infty} A_i = [0, 1]$ and $\cap_{i=1}^{\infty} A_i = \{0\}$. If instead we define $A_i =]0, \frac{1}{i}[$ then $\cup_{i=1}^{\infty} A_i =]0, 1[$ and $\cap_{i=1}^{\infty} A_i = \emptyset$.

4.2 Probability

We will assign a real number $P(A)$ to every event A , called the probability of A . We also call P a **probability distribution** or a **probability measure**. To qualify as a probability, P must satisfy three axioms:

Axiom 1 : $\forall A, P(A) \geq 0$

Axiom 2 : $P(\Omega) = 1$

Axiom 3 : If $A_1, A_2, \dots$ are disjoint then $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

There are many interpretations of $P(A)$. The two common interpretations are frequencies and degrees of beliefs. In the frequency interpretation, $P(A)$ is the long run proportion of times that A is true in repetitions. For example, if we say that the probability of heads is $1/2$, we mean that if we flip the coin many times then the proportion of times we get heads tends to $1/2$ as the number of tosses increases. An infinitely long, unpredictable sequence of tosses whose limiting proportion tends to a constant is an idealization, much like the idea of a straight line in geometry. The degree-of-belief interpretation is that $P(A)$ measures an observer's strength of belief that A is true. In either interpretation, we require that Axioms 1 to 3 hold. The difference in interpretation will not matter much until we deal with statistical inference. There, the differing interpretations lead to two schools of inference: the frequentist and the Bayesian schools.

One can derive many properties of P from the axioms, such as

$$P(\emptyset) = 0$$

$$A \subset B \implies P(A) \leq P(B)$$

$$0 \leq P(A) \leq 1$$

$$P(A^c) = 1 - P(A)$$

$$A \cap B = \emptyset \implies P(A \cup B) = P(A) + P(B)$$

We have the following basic theorem that allows us to decompose the calculation of the probability of B into the sum of the probabilities of the sets $A_i \cap B$. Often these are

easier to compute.

Theorem 4.2.1 (Law of total probability, unconditioned version) *Let $A_1, A_2, \dots$ be events that form a partition of the sample space S . Let B be any event. Then*

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \dots$$

Proof. The events $(A_1 \cap B), (A_2 \cap B), \dots$ are disjoint, and their union is B . Hence, the result follows immediately from the additivity property. ■

Suppose now that A and B are two events such that A contains B (in symbols, $A \supseteq B$). In words, all outcomes in B are also in A . Intuitively, A is a “larger” event than B , so we would expect its probability to be larger. We have the following result.

Theorem 4.2.2 *Let A and B be two events with $A \supseteq B$. Then*

$$P(A) = P(B) + P(A \cap B^c).$$

Proof. We can write $A = B \cup (A \cap B^c)$, where B and $A \cap B^c$ are disjoint. Hence, $P(A) = P(B) + P(A \cap B^c)$ by additivity. ■

A less obvious property is given in the following theorem.

Theorem 4.2.3 (Principle of inclusion–exclusion, two-event version) *For any events A and B ,*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof. Write $A \cup B = (A \cap B^c) \cup (A \cap B) \cup (A^c \cap B)$ and note that these events are disjoint. Hence, making repeated use of the fact that P is additive for disjoint events,

we see that

$$\begin{aligned}
 P(A \cup B) &= P((A \cap B^c) \cup (A \cap B) \cup (A^c \cap B)) \\
 &= P(A \cap B^c) + P(A \cap B) + P(A^c \cap B) \\
 &= P(A \cap B^c) + P(A \cap B) + P(A^c \cap B) + P(A \cap B) - P(A \cap B) \\
 &= P(\underbrace{(A \cap B^c) \cup (A \cap B)}_{=A}) + P(\underbrace{(A^c \cap B) \cup (A \cap B)}_{=B}) - P(A \cap B) \\
 &= P(A) + P(B) - P(A \cap B).
 \end{aligned}$$

■

Example. Two coin tosses. Let H_1 be the event that heads occurs on toss 1 and let H_2 be the event that heads occurs on toss 2. If all outcomes are equally likely, then $P(H_1 \cup H_2) = P(H_1) + P(H_2) - P(H_1 \cap H_2) = \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$.

Theorem 4.2.4 (Continuity of Probabilities) *If $A_n \longrightarrow A$ then*

$$P(A_n) \longrightarrow P(A) \text{ as } n \longrightarrow \infty.$$

Proof. Suppose that A_n is monotone increasing so that $A_1 \subset A_2 \subset \dots$. Let $A = \lim_{n \rightarrow \infty} A_n = \cup_{i=1}^{\infty} A_i$. Define $B_1 = A_1, B_2 = \{\omega \in \Omega : \omega \in A_2, \omega \notin A_1\}, B_3 = \{\omega \in \Omega : \omega \in A_3, \omega \notin A_1, \omega \notin A_2\}, \dots$ It can be shown that $B_1, B_2, \dots$ are disjoint, $\forall n, A_n = \cup_{i=1}^n A_i = \cup_{i=1}^n B_i$ and $\cup_{i=1}^{\infty} A_i = \cup_{i=1}^{\infty} B_i$. From Axiom 3,

$$P(A_n) = P(\cup_{i=1}^n B_i) = \sum_{i=1}^n P(B_i)$$

and hence, using Axiom 3 again,

$$\lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} \sum_{i=1}^n P(B_i) = \sum_{i=1}^{\infty} P(B_i) = P(\cup_{i=1}^{\infty} B) = P(A)$$

■

Sometimes we do not need to evaluate the probability content of a union; we need only know it is bounded above by the sum of the probabilities of the individual events. This is called subadditivity.

Theorem 4.2.5 (Subadditivity) *Let $A_1, A_2, \dots$ be a finite or countably infinite sequence of events, not necessarily disjoint. Then*

$$P(A_1 \cup A_2 \cup \dots) \leq P(A_1) + P(A_2) + \dots$$

4.3 Probability on Finite Sample Spaces

Suppose that the sample space $\Omega = \{\omega_1, \dots, \omega_n\}$ is finite. For example, if we toss a die twice, then Ω has 36 elements: $\Omega = \{(i, j); i, j \in \{1, \dots, 6\}\}$. If each outcome is equally likely, then $P(A) = \frac{|A|}{36}$ where $|A|$ denotes the number of elements in A . The probability that the sum of the dice is 11 is $\frac{2}{36}$ since there are two outcomes that correspond to this event.

If Ω is finite and if each outcome is equally likely, then

$$P(A) = \frac{|A|}{|\Omega|}$$

which is called the **uniform probability distribution**.

Example. Suppose now that we flip three different fair coins. The outcome can be

written as a sequence of three letters, with each letter being H (for heads) or T (for tails). Thus, $\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$. Here $|\Omega| = 8$, and each of the events is equally likely. Hence, $P(\{HHH\}) = 1/8$, $P(\{HHH, TTT\}) = 2/8 = 1/4$, etc. Note also that, by additivity, we have, for example, that $P(\text{exactly two heads}) = P(\{HHT, HTH, THH\}) = 1/8 + 1/8 + 1/8 = 3/8$, etc.

To compute probabilities, we need to count the number of points in an event A . Methods for counting points are called combinatorial methods. We needn't delve into these in any great detail. We will, however, need a few facts from counting theory that will be useful later. Given n objects, the number of ways of ordering these objects is $n! = n(n-1)(n-2)\dots 3 \cdot 2 \cdot 1$. For convenience, we define $0! = 1$. We also define

$$C_n^k = \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

read “ n choose k ”, which is the number of distinct ways of choosing k objects from n . For example, if we have a class of 20 people and we want to select a committee of 3 students, then there are

$$\binom{20}{3} = \frac{20!}{3!17!} = \frac{20 \times 19 \times 18}{3 \times 2 \times 1} = 1140$$

possible committees. We note the following properties:

$$\binom{n}{0} = \binom{n}{n} = 1 \quad \text{and} \quad \binom{n}{k} = \binom{n}{n-k}$$

4.4 Independent Events

If we flip a fair coin twice, then the probability of two heads is $\frac{1}{2} \times \frac{1}{2}$. We multiply the probabilities because we regard the two tosses as independent. The formal definition of independence is as follows:

Définition 4.4.1 *Two events A and B are independent if*

$$P(A \cap B) = P(A)P(B)$$

and we write $A \amalg B$. A set of events $\{A_i : i \in I\}$ is independent if

$$P(\cap_{i \in J} A_i) = \prod_{i \in J} P(A_i)$$

for every finite subset J of I .

Independence can arise in two distinct ways. Sometimes, we explicitly assume that two events are independent. For example, in tossing a coin twice, we usually assume the tosses are independent which reflects the fact that the coin has no memory of the first toss. In other instances, we derive independence by verifying that $P(A \cap B) = P(A)P(B)$ holds. For example, in tossing a fair die, let $A = \{2, 4, 6\}$ and let $B = \{1, 2, 3, 4\}$. Then, $A \cap B = \{2, 4\}$, $P(A \cap B) = 2/6 = P(A)P(B) = (1/2) \times (2/3)$ and so A and B are independent. In this case, we didn't assume that A and B are independent — it just turned out that they were.

Suppose that A and B are disjoint events, each with positive probability. Can they be independent? No. This follows since $P(A)P(B) > 0$ yet $P(A \cap B) = P(\emptyset) = 0$.

Example. Toss a fair coin 10 times. Let A = “at least one head.” Let T_j be the event that tails occurs on the j^{th} toss. Then

$$\begin{aligned}
 P(A) &= 1 - P(A^c) \\
 &= 1 - P(\text{all tails}) \\
 &= 1 - P(T_1 \cap T_2 \cap \cdots \cap T_{10}) \\
 &= 1 - P(T_1)P(T_2) \cdots P(T_{10}) \text{ using independence} \\
 &= 1 - \left(\frac{1}{2}\right)^{10} \approx 0.999
 \end{aligned}$$

Example. Two people take turns trying to sink a basketball into a net. Person 1 succeeds with probability $1/3$ while person 2 succeeds with probability $1/4$. What is the probability that person 1 succeeds before person 2?

Let E denote the event of interest. Let A_j be the event that the first success is by person 1 and that it occurs on trial number j . Note that $A_1, A_2, \dots$ are disjoint and that $E = \cup_{j=1}^{\infty} A_j$. Hence,

$$P(E) = \sum_{j=1}^{\infty} P(A_j)$$

Now, $P(A_1) = 1/3$. A_2 occurs if we have the sequence person 1 misses, person 2 misses, person 1 succeeds. This has probability $P(A_2) = (2/3)(3/4)(1/3) = (1/2)(1/3)$. Following this logic we see that $P(A_j) = (1/2)^{j-1}(1/3)$. Hence,

$$P(E) = \sum_{j=1}^{\infty} \left(\frac{1}{2}\right)^{j-1} \frac{1}{3} = \frac{1}{3} \sum_{j=1}^{\infty} \left(\frac{1}{2}\right)^{j-1} = \frac{1}{3} \frac{1}{1 - \frac{1}{2}} = \frac{2}{3}.$$

Here we used that fact that, if $0 < r < 1$ then $\sum_{j=k}^{\infty} r^j = \frac{r^k}{1-r}$.

4.5 Conditional Probability

Assuming that $P(B) > 0$, we define the conditional probability of A given that B has occurred as follows:

Définition 4.5.1 *If $P(B) > 0$ then the conditional probability of A given B is*

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Think of $P(A|B)$ as the fraction of times A occurs among those in which B occurs. For any fixed B such that $P(B) > 0$, $P(\cdot|B)$ is a probability (i.e., it satisfies the three axioms of probability). In particular, $P(A|B) \geq 0$, $P(\Omega|B) = 1$ and if $A_1, A_2, \dots$ are disjoint then $P(\cup_{i=1}^{\infty} A_i|B) = \sum_{i=1}^{\infty} P(A_i|B)$. But it is in general **not** true that $P(A|B \cup C) = P(A|B) + P(A|C)$. The rules of probability apply to events on the left of the bar.

In general it is not the case that $P(A|B) = P(B|A)$. People get this confused all the time. For example, the probability of spots given you have measles is 1 but the probability that you have measles given that you have spots is not 1. In this case, the difference between $P(A|B)$ and $P(B|A)$ is obvious but there are cases where it is less obvious.

Example. A medical test for a disease D has outcomes $+$ and $-$. The probabilities are:

	D	D^c
$+$	0.009	0.099
$-$	0.001	0.891

From the definition of conditional probability,

$$P(+|D) = \frac{P(+ \cap D)}{P(D)} = \frac{0.009}{0.009 + .001} = 0.9$$

and

$$P(-|D^c) = \frac{P(- \cap D^c)}{P(D^c)} = \frac{0.891}{0.891 + 0.099} \approx 0.9.$$

Apparently, the test is fairly accurate. Sick people yield a positive 90 percent of the time and healthy people yield a negative about 90 percent of the time.

Suppose you go for a test and get a positive. What is the probability you have the disease? Most people answer 0.9. The correct answer is

$$P(D|+) = \frac{P(+ \cap D)}{P(+)} = \frac{0.009}{0.009 + 0.099} \approx 0.08.$$

The lesson here is that you need to compute the answer numerically. Don't trust your intuition.

Proposition 4.5.1 (Compound probability formula) *Let n events $A_1, \dots, A_n$ be such that $P(A_1 \cap \dots \cap A_n) \neq 0$. Then*

$$P(A_1 \cap \dots \cap A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \dots P(A_n|A_1 \cap A_2 \dots \cap A_{n-1}).$$

Example. An urn initially contains 7 black balls and 3 white balls. We successively draw 3 balls: if we draw a black one, we remove it, if we draw a white one, we remove it, and we add a black one instead. What is the probability of drawing 3 blanks in a row?

We denote by B_i the event "The i -th ball drawn is white". The desired probability is

$$P(B_1 \cap B_2 \cap B_3) = P(B_1)P(B_2|B_1)P(B_3|B_1 \cap B_2).$$

Clearly, $P(B_1) = 3/10$. Now, if B_1 is made, before the 2nd draw, the urn consists of 8 black and 2 white balls. We therefore have: $P(B_2|B_1) = 2/10$. If B_1 and B_2 are made, before the 3rd draw, the urn consists of 9 black balls and 1 white. We deduce $P(B_3|B_1 \cap B_2) = 1/10$. Finally $P(B_1 \cap B_2 \cap B_3) = 6/1000 = 3/500$.

4.6 Bayes' Theorem

First, we need a preliminary result. This formula makes it possible to calculate the probability of an event B by breaking it down according to a complete system of events (Indeed, B is equal to the disjoint union of $B \cap A_n$)

Theorem 4.6.1 (The Law of Total Probability) *Let $A_1, \dots, A_k$ be a partition of Ω . Then, for any event B ,*

$$P(B) = \sum_{i=1}^k P(B|A_i)P(A_i).$$

Proof. Define $C_j = B \cap A_j$ and note that $C_1, \dots, C_k$ are disjoint and that $B = \cup_{j=1}^k C_j$. Hence,

$$P(B) = P(\cup_{j=1}^k C_j) = \sum_{j=1}^k P(B \cap A_j) = \sum_{j=1}^k P(B|A_j)P(A_j)$$

since $P(B \cap A_j) = P(B|A_j)P(A_j)$ from the definition of conditional probability. ■

Theorem 4.6.2 (Bayes' Theorem) *Let $A_1, \dots, A_k$ be a partition of Ω such that $P(A_i) >$*

0 for each i . If $P(B) > 0$ then, for each $i = 1, \dots, k$,

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^k P(B|A_j)P(A_j)} .$$

Proof. We apply the definition of conditional probability twice, followed by the law of total probability:

$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)} = \frac{P(B|A_i)P(A_i)}{P(B)} = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^k P(B|A_j)P(A_j)} .$$

■

Example. We divide emails into three categories: A_1 = “spam,” A_2 = “low priority” and A_3 = “high priority.” From previous experience we find that $P(A_1) = 0.7$, $P(A_2) = 0.2$ and $P(A_3) = 0.1$. Of course, $0.7 + 0.2 + 0.1 = 1$. Let B be the event that the email contains the word “free”. From previous experience, $P(B|A_1) = 0.9$, $P(B|A_2) = 0.01$, $P(B|A_3) = 0.01$. (Note: $0.9 + 0.01 + 0.01 \neq 1$). We receive an email with the word “free”. What is the probability that it is spam?

Bayes’ theorem yields,

$$P(A_1|B) = \frac{0.9 \times 0.7}{(0.9 \times 0.7) + (0.01 \times 0.2) + (.001 \times 0.1)} = 0.995.$$

4.7 Random Variables and Distribution

In previous sections, we discussed the probability model as the central object of study in the theory of probability. This required defining a probability measure P on a class of subsets of the sample space Ω . It turns out that there are simpler ways of presenting a particular probability assignment than this — ways that are much more convenient

to work with than P . This section is concerned with the definitions of random variables, distribution functions, probability functions, and the development of the concepts necessary for carrying out calculations for a probability model using these entities. This section also discusses the concept of the conditional distribution of one random variable, given the values of others. Conditional distributions of random variables provide the framework for discussing what it means to say that variables are related, which is important in many applications of probability and statistics.

4.7.1 Random Variables

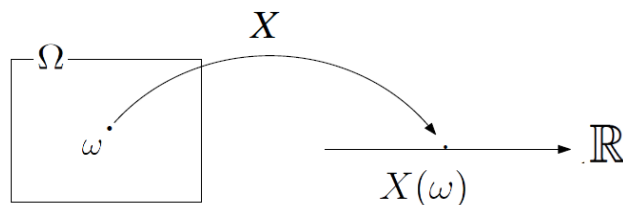
The previous sections explained how to construct probability models, including a sample space Ω and a probability measure P . Once we have a probability model, we may define *random variables* for that probability model.

Intuitively, a random variable assigns a numerical value to each possible outcome in the sample space. For example, if the sample space is $\{\text{rain, snow, clear}\}$, then we might define a random variable X such that $X = 3$ if it rains, $X = 6$ if it snows, and $X = -2.7$ if it is clear.

More formally, we have the following definition.

Définition 4.7.1 *A random variable is a function from the sample space Ω to the set $\mathbb{R}$ of all real numbers that assigns a real number $X(\omega)$ to each outcome ω .*

$$\begin{aligned} X : \Omega &\longrightarrow \mathbb{R} \\ \omega &\longmapsto X(\omega) = x \end{aligned}$$



The random variable described above could be written formally as $X : \{\text{rain, snow, clear}\} \rightarrow \mathbb{R}$ by $X(\text{rain}) = 3$, $X(\text{snow}) = 6$, and $X(\text{clear}) = -2.7$. We will return to this example below.

We now present several further examples. The point is, we can define random variables any way we like, as long as they are functions from the sample space to $\mathbb{R}$.

Example. If the sample space corresponds to flipping three different coins, then we could let X be the total number of heads showing, let Y be the total number of tails showing, let $Z = 0$ if there is exactly one head, and otherwise $Z = 17$, etc.

Example. If the sample space corresponds to rolling two fair dice, then we could let X be the square of the number showing on the first die, let Y be the square of the number showing on the second die, let Z be the sum of the two numbers showing, let W be the square of the sum of the two numbers showing, let R be the sum of the squares of the two numbers showing, etc.

Example. (Constants as Random Variables) As a special case, every constant value c is also a random variable, by saying that $c(\omega) = c$ for all $\omega \in \Omega$. Thus, 5 is a random variable, as is 3 or -21.6 .

Example. (Indicator Functions) One special kind of random variable is worth mentioning. If A is any event, then we can define the *indicator function* of A , written $\mathbb{I}_A$, to be the random variable

$$\mathbb{I}_A(\omega) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}$$

which is equal to 1 on A , and is equal to 0 on A^c .

Given random variables X and Y , we can perform the usual arithmetic operations on them. Thus, for example, $Z = X^2$ is another random variable, defined by $Z(\omega) = X^2(\omega) = (X(\omega))^2 = X(\omega) \times X(\omega)$. Similarly, if $W = XY^3$, then $W(\omega) = X(\omega) \times Y(\omega) \times Y(\omega) \times Y(\omega)$, etc. Also, if $Z = X + Y$, then $Z(\omega) = X(\omega) + Y(\omega)$, etc.

Example. Consider rolling a fair six-sided die, so that $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let X be the number showing, so that $X(\omega) = \omega$ for $\omega \in \Omega$. Let Y be three more than the number showing, so that $Y(\omega) = \omega + 3$. Let $Z = X^2 + Y$. Then $Z(\omega) = X(\omega)^2 + Y(\omega) = \omega^2 + \omega + 3$. So $Z(1) = 5$, $Z(2) = 9$, etc.

We write $X = Y$ to mean that $X(\omega) = Y(\omega)$ for all $\omega \in \Omega$. Similarly, we write $X \leq Y$ to mean that $X(\omega) \leq Y(\omega)$ for all $\omega \in \Omega$, and $X \geq Y$ to mean that $X(\omega) \geq Y(\omega)$ for all $\omega \in \Omega$. For example, we write $X \leq c$ to mean that $X(\omega) \leq c$ for all $\omega \in \Omega$.

Example. Again consider rolling a fair six-sided die, with $\Omega = \{1, 2, 3, 4, 5, 6\}$. For $\omega \in \Omega$, let $X(\omega) = \omega$, and let $Y = X + \mathbb{I}_{\{6\}}$. This means that

$$Y(\omega) = X(\omega) + \mathbb{I}_{\{6\}}(\omega) = \begin{cases} \omega & \omega \leq 5 \\ 7 & \omega = 6 \end{cases}$$

If S is infinite, then a random variable X can take on infinitely many different values.

Example. If $\Omega = \{1, 2, 3, \dots\}$, if X is defined by $X(\omega) = \omega^2$, then we always have $X \geq 1$. But there is no largest value of $X(\omega)$ because the value $X(\omega)$ increases without bound as $\omega \rightarrow \infty$. We shall call such a random variable an *unbounded random variable*. This is pretty much the same as a statistical variable except that in the case of a statistical variable one evaluates a realized behavior (average, etc) whereas in the case of random variables one assumes a future behavior (In this case, we speak of expectation

rather than average for example) or theoretical.

Random variables are used to model the outcome of a non-deterministic mechanism.

Définition 4.7.2 *A random variable (or r.v.) is a map $X : \Omega \rightarrow \mathbb{R}$. If $X(\Omega)$ is at most countable, we say that X is a **discrete** r.v. otherwise we say that it is **continuous**.*

Finally, suppose X is a random variable. We know that different states ω occur with different probabilities. It follows that $X(\omega)$ also takes different values with different probabilities. These probabilities are called the *distribution of X* ; we consider them next.

4.7.2 Distributions of Random Variables

Because random variables are defined to be functions of the outcome ω , and because the outcome ω is assumed to be random (i.e., to take on different values with different probabilities), it follows that the value of a random variable will itself be random (as the name implies).

Specifically, if X is a random variable, then what is the probability that X will equal some particular value x ? Well, $X = x$ precisely when the outcome ω is chosen such that $X(\omega) = x$.

A random variable is totally defined by its law of probability and it is characterized by:

- the set of values it can take (its domain of definition);
- the probabilities attributed to each of the potentially taken values $P(X = x)$.

In this case, the law of the random variable is the law of probability on the set of possible values of X which assigns the probability $P(X = x)$ to the singleton $\{x\}$.

Example. Let us again consider the random variable of the first example above, where $\Omega = \{\text{rain, snow, clear}\}$, and X is defined by $X(\text{rain}) = 3$, $X(\text{snow}) = 6$, and

$X(\text{clear}) = -2.7$. Suppose further that the probability measure P is such that $P(\text{rain}) = 0.4$, $P(\text{snow}) = 0.15$, and $P(\text{clear}) = 0.45$. Then clearly, $X = 3$ only when it rains, $X = 6$ only when it snows, and $X = -2.7$ only when it is clear. Thus, $P(X = 3) = P(\text{rain}) = 0.4$, $P(X = 6) = P(\text{snow}) = 0.15$, and $P(X = -2.7) = P(\text{clear}) = 0.45$. Also, $P(X = 17) = 0$, and in fact $P(X = x) = P(\emptyset) = 0$ for all $x \notin \{3, 6, -2.7\}$. We can also compute that

$$P(X \in \{3, 6\}) = P(X = 3) + P(X = 6) = 0.4 + 0.15 = 0.55,$$

while

$$P(X < 5) = P(X = 3) + P(X = -2.7) = 0.4 + 0.45 = 0.85,$$

etc.

4.7.3 Distribution function

Définition 4.7.3 (cumulative distribution function) *The distribution function (CDF) of an r.v. X is the map F_X of $\mathbb{R}$ in $[0, 1]$ defined by*

$$F_X(x) = P(X \leq x) = P(X^{-1}([-\infty, x]))$$

We are often interested in the cumulative probability. For example in the case of probabilities over $\mathbb{N}$:

$$P(X \leq n) = P(X = 0 \text{ or } X = 1 \text{ or } \dots \text{ or } X = n).$$

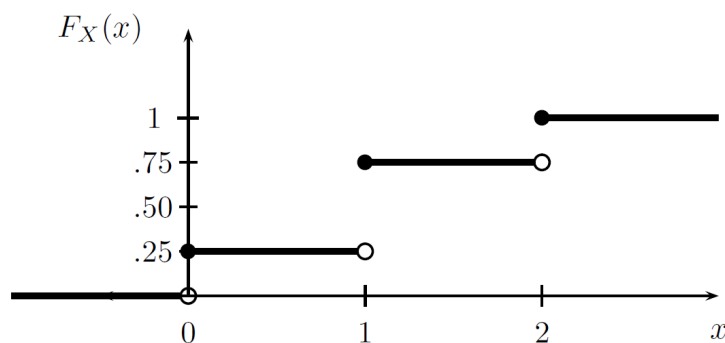
Since the events are mutually exclusive, we obtain

$$P(X \leq n) = \sum_{j=0}^n P(X = j).$$

We will see later that the CDF effectively contains all the information about the random variable. Sometimes we write the CDF as F instead of F_X .

Example. Flip a fair coin twice and let X be the number of heads. Then $P(X = 0) = P(X = 2) = 1/4$ and $P(X = 1) = 1/2$. The distribution function is

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{4} & 0 \leq x < 1 \\ \frac{3}{4} & 1 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$



The CDF is shown in the above figure. Although this example is simple, study it carefully. CDF's can be very confusing. Notice that the function is right continuous, non-decreasing, and that it is defined for all x , even though the random variable only takes values 0, 1, and 2. Do you see why $F_X(1.4) = 0.75$?

The following result shows that the CDF completely determines the distribution of a random variable.

Theorem 4.7.1 *Let X have CDF F and let Y have CDF G . If $F(x) = G(x)$ for all x , then $P(X \in A) = P(Y \in A)$ for all A .*

In fact, we say that the two random variables X and Y are **equal in distribution** — written $X \stackrel{d}{=} Y$ — if $F_X(x) = F_Y(x)$ for all x and this does not mean that X and Y are equal. Rather, it means that all probability statements about X and Y will be the same. For example, suppose that $P(X = 1) = P(X = -1) = 1/2$. Let $Y = -X$. Then $P(Y = 1) = P(Y = -1) = 1/2$ and so $X \stackrel{d}{=} Y$. But X and Y are not equal. In fact, $P(X = Y) = 0$.

Theorem 4.7.2 *A function F mapping the real line to $[0, 1]$ is a CDF for some probability P if and only if F satisfies the following three conditions:*

- (i) *F is non-decreasing: $x_1 < x_2$ implies that $F(x_1) \leq F(x_2)$.*
- (ii) *F is normalized: $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.*
- (iii) *F is right-continuous: $F(x) = F(x^+)$ for all x , where $F(x^+) = \lim_{y \searrow x} F(y)$.*

Proof. (i) Since $x_1 < x_2$, then we have $] - \infty, x_1] \subset] - \infty, x_2]$ which gives

$$P(X \in] - \infty, x_1]) \leq P(X \in] - \infty, x_2]) \iff F(x_1) \leq F(x_2)$$

(ii) Obvious

(iii) Suppose that F is a CDF. Let x be a real number and let $y_1, y_2, \dots$ be a sequence of real numbers such that $y_1 > y_2 > \dots$ and $y_i \xrightarrow{i \rightarrow \infty} x$. Let $A_i =] - \infty, y_i]$ and let $A =] - \infty, x]$. Note that $A = \cap_{i=1}^{\infty} A_i$ and also note that $A_1 \supset A_2 \supset \dots$. Because the events are monotone, $\lim_{i \rightarrow \infty} P(A_i) = P(\cap_{i=1}^{\infty} A_i)$. Thus,

$$F(x) = P(A) = P(\cap_{i=1}^{\infty} A_i) = \lim_{i \rightarrow \infty} P(A_i) = \lim_{i \rightarrow \infty} F(y_i) = F(x^+).$$

■

Lemma 4.7.1 *Let F be the CDF for a random variable X . Then:*

$$(i) P(x_1 < X \leq x_2) = F(x_2) - F(x_1)$$

$$(ii) P(X > x) = 1 - F(x).$$

Proof. (i) Take $x_1 < x_2$, then

$$\begin{aligned} P(X \leq x_2) &= P(X \in]-\infty, x_2]) = P(\{X \in]-\infty, x_1]\} \cup \{X \in]x_1, x_2]\}) \\ &= P(X \in]-\infty, x_1]) + P(X \in]x_1, x_2]) \\ &= P(X \leq x_1) + P(x_1 < X \leq x_2) \\ &\implies P(x_1 < X \leq x_2) = P(X \leq x_2) - P(X \leq x_1) = F(x_2) - F(x_1) \end{aligned}$$

(ii) We replace x_2 in (i) by ∞ . ■

4.7.4 Expectation and variance of a random variable

Average value of a random variable: Expectation

The intuitive idea of expectation has its origins in games of chance. Consider the following game: a die is rolled. Suppose that for a bet of 1 euro, we win 1 euro if the result obtained is even, 2 euros if the result is 1 or 3, and we lose 3 euros if the result is 5. Is it interesting to play this game? What is the average gain?

Let X be the random variable corresponding to the number of euros won or lost. The law of X is

x	-3	1	2
$P(X = x)$	1/6	1/2	1/3

The expected win, denoted $E[X]$, is then $E[X] = -3 \times 1/6 + 1 \times 1/2 + 2 \times 1/3 = 2/3$.

The player therefore earns on average $2/3$ euros for a bet of 1 euro.

An *expectation operator* is a function that assigns to each random variable X a real number $E[X]$ called the expectation or expected value of X .

Every expectation operator satisfies the following axioms.

Axiom E1 (Additivity). If X and Y are random variables, then $X + Y$ is also a random variable, and

$$E[X + Y] = E[X] + E[Y].$$

Axiom E2 (Homogeneity). If X is random variable and a is a real number, then aX is also a random variable, and

$$E[aX] = aE[X].$$

These properties agree with either of the informal intuitions about expectations. Prices are additive and homogeneous. The price of a gallon of milk and a box of cereal is the sum of the prices of the two items separately. Also the price of three boxes of cereal is three times the price of one box. (The notion of expectation as fair price doesn't allow for volume discounts.)

Axiom E3 (Positivity). If X is random variable, then $X \geq 0$ implies $E[X] \geq 0$.

The expression $X \geq 0$, written out in more detail, means

$$X(\omega) \geq 0, \omega \in \Omega,$$

where Ω is the sample space. That is, X is always nonnegative. This axiom corresponds to intuition about prices, since goods always have nonnegative value and prices are also

nonnegative.

Définition 4.7.4 *The expectation of a random variable X is denoted $E[X]$. It represents the average value taken by the variable X .*

1. *If X is a discrete variable with values in $D = \{x_1, \dots, x_n\}$, its expectation is*

$$E[X] = x_1P(X = x_1) + \dots + x_nP(X = x_n) = \sum_{i=1}^n x_iP(X = x_i)$$

2. *If X is a discrete variable with values in the infinite set $D = \{x_i : i \geq 1\}$, when the sum is well defined, its expectation is*

$$E[X] = \sum_{i=1}^{\infty} x_iP(X = x_i)$$

When a variable X verifies $E[X] = 0$, we say that the variable is **centered**.

Proposition 4.7.1 1. *the expectation is linear: let a and $b \in \mathbb{R}$ and two random variables X and Y with finite expectation then*

$$E[aX + bY] = aE[X] + bE[Y].$$

2. *If $X \leq Y$ then $E[X] \leq E[Y]$.*

The Multiplicativity Non-Property

One might suppose that there is a property analogous to the additivity property, except with multiplication instead of addition

$$E[XY] = E[X]E[Y], \text{ Uncorrelated } X \text{ and } Y \text{ only!}$$

As the editorial comment says, this property does not hold in general. We will later see that when it does hold we have a special name for this situation: we say the variables X and Y are uncorrelated.

Taking a Function Outside an Expectation

Suppose g is a linear function defined by

$$g(x) = a + bx, x \in \mathbb{R},$$

where a and b are real numbers. Then

$$E[g(X)] = g(E[X]), \text{ Linear } g \text{ only!}$$

The reason for the editorial comment is that property does not hold for general functions g , only for linear functions. Sometime you will be tempted to use it for a nonlinear function g . Don't! Remember that it is a “non-property.”

For example, you may be asked to calculate $E[1/X]$ for some random variable X . The “non-property”, if it were true, would allow to take the function outside the expectation and the answer would be $1/E[X]$, but it isn't true, and, in general

$$E\left[\frac{1}{X}\right] \neq \frac{1}{E[X]}.$$

4.7.5 Moments

If k is a positive integer, then the real number

$$\alpha_k = E[X^k]$$

is called the $k - th$ moment of the random variable X .

If p is a positive real number, then the real number

$$\beta_p = E[|X|^p]$$

is called the $p - th$ absolute moment of the random variable X .

If k is a positive integer and $\mu = E[X]$, then the real number

$$\mu_k = E[(X - \mu)^k]$$

is called the $k - th$ central moment of the random variable X .

That's not the whole story on moments. We can define lots more, but all moments are special cases of one of the two following concepts.

If k is a positive real number and a is any real number, then the real number $E[(X - a)^k]$ is called the $k - th$ moment about the point a of the random variable X . We introduce no special symbol for this concept. Note that the $k - th$ ordinary moment is the special case $a = 0$ and the $k - th$ central moment is the case $a = \mu$.

If p is a positive real number and a is any real number, then the real number $E[|X - a|^p]$ is called the $p - th$ absolute moment about the point a of the random variable X . Note that the $p - th$ absolute moment is the special case $a = 0$.

The first ordinary moment of a random variable X is also called the mean of X . It is commonly denoted by the Greek letter μ . Note that α_1 , μ , and $E[X]$ are different notations for the same thing. We will use them all throughout the course.

When there are several random variables under discussion, we denote the mean of each using the same Greek letter μ , but add the variable as a subscript to distinguish them: $\mu_X = E[X]$, $\mu_Y = E[Y]$, and so forth.

Theorem 4.7.3 *Suppose a real-valued random variable X is symmetric about the point a . If the mean of X exists, it is equal to a . Every higher odd integer central moment of X that exists is zero*

In notation, the two assertions of the theorem are

$$E[X] = \mu = a$$

and

$$\mu_{2k+1} = E[(X - \mu)^{2k+1}] = 0, \text{ for any positive integer } k.$$

The proof is left as an exercise.

Second Moments and Variances

The preceding section says all that can be said in general about first moments. As we shall now see, second moments are much more complicated.

The most important second moment is the second central moment, which also has a special name. It is called the *variance* and is often denoted σ^2 . We will see the reason for the square presently. We also use the notation $\text{var}(X)$ for the variance of X . So

$$\sigma^2 = \mu_2 = \text{var}(X) = E[(X - \mu)^2].$$

As we did with means, when there are several random variables under discussion, we denote the variance of each using the same Greek letter σ , but add the variable as a subscript to distinguish them: $\sigma_X^2 = \text{var}(X)$, $\sigma_Y^2 = \text{var}(Y)$, and so forth.

Note that variance is just an expectation like any other, the expectation of the random variable $(X - \mu)^2$.

Corollary 4.7.1 *If X is a random variable having first and second moments, then*

$$\text{var}(X) = E[X^2] - E[X]^2.$$

There are various ways of restating the corollary in symbols, for example

$$\sigma_X^2 = E[X^2] - \mu_X^2$$

and

$$\mu_2 = \alpha_2 - \alpha_1^2$$

As always, mathematics is invariant under changes of notation. The important thing is the concepts symbolized rather than the symbols themselves.

Theorem 4.7.4 *Suppose X is a random variable having first and second moments and a and b are real numbers, then*

$$\text{var}(a + bX) = b^2 \text{var}(X)$$

Note that the right hand side does not involve the constant part a of the linear transformation $a + bX$. Also note that the b comes out squared.

The nonnegative square root of the variance is called the *standard deviation*. Conversely, the variance is the square of the standard deviation. The symbol commonly used for the standard deviation is σ . That's why the variance is usually denoted σ^2 .

As with the mean and variance, we use subscripts to distinguish variables σ_X , σ_Y , and so forth.

It might have just occurred to you to ask why anyone would want two such closely re-

lated concepts. Won't one do? In fact more than one introductory (freshman level) statistics textbook does just that, speaking only of standard deviations, never of variances. But for theoretical probability and statistics, this will not do. Standard deviations are almost useless for theoretical purposes. The square root introduces nasty complications into simple situations. So for theoretical purposes variance is the preferred concept. In contrast, for all practical purposes standard deviation is the preferred concept, as evidenced by the fact that introductory statistics textbooks that choose to use only one of the two concepts invariably choose standard deviation. The reason has to do with units of measurement and measurement scales. Suppose we have a random variable X whose units of measurement are inches, for example, the height of a student in the class. What are the units of $E[X]$, $var(X)$, and σ_X , assuming these quantities exist?